Introduction to Big Data & IoT

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What is Big Data?

Big Data helps businesses make better use of their data.

- Things customers write on social media
- Log files from applications
- Processes
- Sensor on machines
- Device data (smart phones, laptops, etc.)

Big Data represents a huge opportunity as a source of competitive advantage

What is the Internet of Things (IoT)?

Collect all data and analyze it in real time to find important trends or patterns.

Three primary objectives of IoT:

1. Provide insights into how we can improve processes, products, and quality.
2. Predict failures before they happen to minimize downtime.
3. Reduce costs by optimizing equipment and maximizing resources.

One way to think of it is a one of the sources of big data.

What is the difference between Big Data and the Internet of Things?

Data Analytics

Data Analytics is fundamentally about generating new insights on data in many formats from many sources.

Data created by sensors, devices, and products, in combination with many other sources, is invaluable in the above three scenarios to inform decisions related to:

- Demand forecasting
- Product launches
- Research and Development
- Customer Satisfaction and Advocacy
- Cross or Upsell
- Personalized Offers
- On Shelf Availability

These are analytical use cases that can work without IoT data, but using IoT data makes them much better.

Big Data and IoT together generate much more data for powerful data analytics.

Big Data and the IoT: a powerful combination

Big data and the Internet of Things (IoT) are two of the most exciting developments in the business world. Each is significant on its own, but together they offer even greater potential. IoT plays a crucial role in the larger discussion surrounding big data analytics.

The value is in the data.

Big Data and IoT Address Challenges

- Reducing overall maintenance costs.
- Ensuring consistent product quality and complying with standards.
- Getting real-time notifications when the production yield or quality changes.
- Getting real-time notifications before your equipment fails.
- Predicting equipment outages.
- Integrating your factory floor with your enterprise applications.

IoT Use Cases

1. A courier service could use IoT data to monitor its drivers and vehicles, including routes followed, fuel used, and even the temperature in the cargo bins when sensitive materials are transported. Tracking these data points would allow the company to improve efficiency and performance, as well as to ensure optimum conditions for each shipment.

2. A sporting goods manufacturer could put sensors in its rackets and bats to allow athletes to analyze ball speed, spin, and impact location.

3. A retail chain could use IoT data to track customer locations, tailor unique offers, and boost participation in its loyalty program.

4. A heavy equipment manufacturer can install sensors in its machines to warn technicians when processes are not working correctly. Then, they can analyze the data to forecast when maintenance will be needed.

Real World IoT and Big Data Platforms

Oracle's IoT and Big Data platforms work together and complement each other.

- Securely connect and acquire data from any device
- Analyze in-flight data and drive insights from IoT analytics
- Integrate IoT data into your enterprise applications
- Create a cost-effective, scalable, open architecture for big data analytics in the cloud
- Automate lifecycle management to run a wide variety of big data workloads
- Analyze data across Hadoop, NoSQL, and Oracle Database Service - Exadata Edition
- Enable in-depth analytics using Oracle Advanced Analytics and Oracle R Advanced Analytics for Hadoop

Summary

Big Data analytics over vast volumes of data, correlating sensor signals with historical and reference data using big data technologies.

IoT defines the logic to process data, define application services, and manage the data ingestion process. You can also define multiple data sources, run ad-hoc queries, package and deploy your applications, and monitor live applications.

Internet of Things allows you to integrate sensors and devices with very different characteristics (such as protocol, data formats, and communication protocols).

Questions